Readers do commit incrementally to causal inferences, under pressure for rich encoding

John Duff (UCLA Ling, duff@ucla.edu), Pranav Anand & Amanda Rysling (UCSC Ling)

While representations of meaning seem to be constructed during incremental comprehension, it can be hard to find reading time evidence for this, when compared to the well-established effects of real-time syntactic processing. This is especially true for pragmatic inferences across large spans of text, for which studies have found contradictory evidence: e.g., [6] reported an eye movement study in which relative clauses that provided plausible explanations for a recently-described event lead to difficulty on a following explicit explanation, consistent with the hypothesis that readers make rapid causal inferences, which may then be retracted. But [4] probed for similar inferences in a highly-powered SPR study, and failed to find the same effect.

Part of the difficulty might be that SPR is notoriously noisy and prone to shallow interpretation, and effects of incremental meaning-building may be hard to find under these conditions, as has been argued for real-time aspectual interpretation [8]. Alternative measures, like word-by-word decision latencies from the Maze task, may alleviate measurement noise and encourage more close interpretation [5]. While the Maze has been effective in recovering highly-local interpretative effects [7], [4]'s Maze results also revealed no effects of causal inferencing; see also [10], where neither SPR nor Maze supported effects of temporal iconicity inferences [3].

In the present pre-registered study [P], we revisited [4]'s materials under a task manipulation argued to support rich incremental encoding of discourse-level meaning. Only in this case did we observe evidence for costly retraction of a causal inference. We consider the implications of this task-specificity, and advocate for this design as a best practice in discourse processing.

[2] introduced the **Reading + Arithmetic + Comprehension Question (R/A/CQ)** task in a study probing the incremental interpretation of definite descriptions in story contexts. Initial SPR and eye movement studies found no incremental cost associated with definites like "the stove" in story contexts when they would not be unique ("the appliance store"), and therefore expected to provoke difficulty. But the expected cost emerged in other studies where readers were asked to complete a simple math problem between each passage and its CQ. [2, cf. 9] suggested that the presence of the intervening math encouraged readers to construct representations of discourse meaning which were richer and more stable than typically required for reading + CQs.

Our study applied the same task with [4]'s items. 80 participants saw 40 critical six-sentence stories in English (1), in which S1+S2 set up a choice scenario for a protagonist, S3 listed their choice *C*, S4 provided a salient feature *F* of the chosen option which would make it appealing, but S6 later revealed that the protagonist's choice was actually driven by an alternative feature. S2 was manipulated across two conditions, such that the protagonist was **KNOW**ledgeable or **IGNO**rant of *F*—the critical causal inference that they chose *C* due to *F* is expected to be more likely in KNOW. Participants completed large-chunked SPR, among 20 fillers where similar causal inferences were not retracted and 50 unrelated fillers. Every passage was followed by a math question (2) and two-alternative CQ (3), both followed by feedback to encourage attention.

Results: S6 was read ~100ms slower in KNOW contexts, consistent with a rapid causal inference that was retracted at cost, contrasting with no effect in [4]'s word-by-word SPR and Maze latencies. The effect did not vary across the duration of the study. CQ accuracy patterns (Fig. 4) provide further suggestion that this task encouraged inferencing: while accuracy did not generally decrease compared to other tasks, questions targeting the erroneous causal inference (S6) in KNOW contexts fell from 85% accuracy in [4]'s SPR data to 65% here. We also note that the critical RT index featured much less variance when using large-chunked SPR vs. [4]'s summing across multiple regions (Fig. 2), improving our power to detect such RT effects [1].

Our results provide converging evidence that readers can enrich representations of textual meaning with inferences during incremental comprehension. The task-dependence of this effect suggests that readers do not always develop such representations in experimental contexts, as in [2], and that task pressures for persistent representations (and chunks matched to analytical regions) are a particularly useful tool for combating this reluctance.

(1) S1: Anna had to edit a bunch of videos for work /
and went to the store to buy a new laptop so she could do it easily. /
S2: (IGNO) She doesn't know a lot about computers and she didn't do much research. /
(KNOW) She carefully researched the specifications for each laptop currently on sale. /
S3: In the end, she went for the MagicBook. /
S4: It has the most powerful processor on the market today. /
S5: It performs far better than its competitors on most tests. /
S6: She bought it because it has a lifetime warranty.

(2) **Arithmetic question form:** XX {+ or –} X = __.
Participants typed their answers before proceeding.

(3) **Example comprehension questions, identical to [4]:**
(a) Targeting protagonist knowledge ("S2," ⅙ of CQs):
Did Patricia research dentists before her appointment? (Yes, No)
(b) Targeting reason for choice ("S6," ⅙ of CQs):
Why did Sasha pick Global Catering? (Their desserts, Their tacos)

(4) **Model call for RT analyses below**:

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brm(RT ~ NWords + Knowledge * Trial + (1 + NWords + Knowledge * Trial | Subject) +
    (1 + Knowledge | Item), family = lognormal()...)
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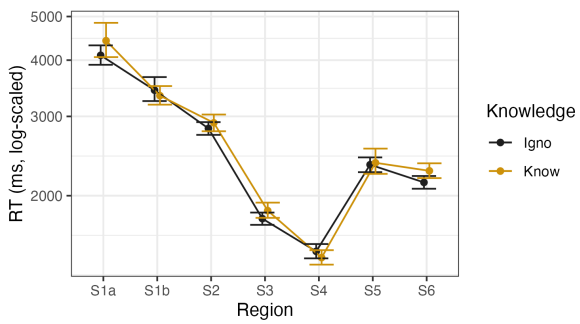

Figure 1: Average and bootstrapped 95% CIs by region. (No other regions display credible effects.)

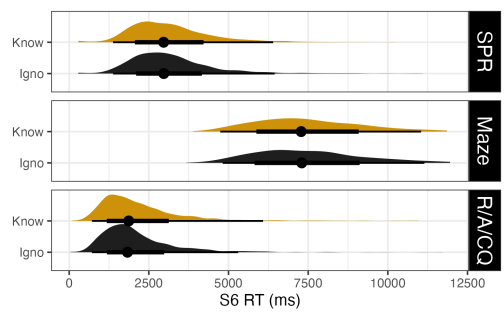

Figure 2: Distributions of S6 RTs across the three studies.

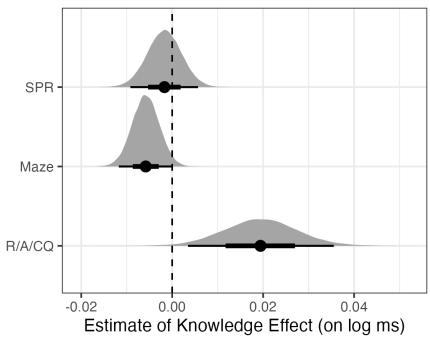

Figure 3: Posterior $\hat{\beta}_{\text{Know1}}$ across the three studies.

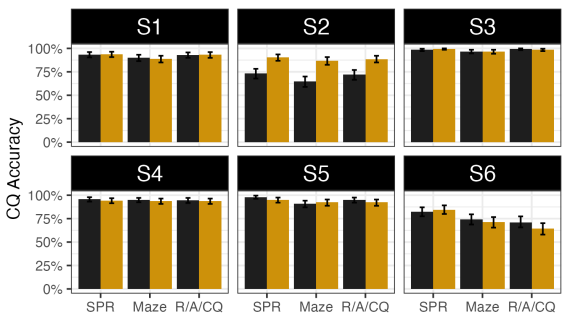

Figure 4: CQ results by question type in the three studies.

References: [1] Almor & Nair (2007) *Lang Ling Comp* [2] Clifton (2013) *JEP:LMC* [3] Dickey (2001) Kluwer [4] Duff (2023) UC Santa Cruz Diss. [5] Forster et al. (2009) *Behav Res Meth* [6] Hoek et al. (2021) *Cogn* [7] Levinson (2023) *Proceedings of ELM* [8] Pickering et al. (2006) *Disc Proc* [9] D. Rohde (2003) at CUNY 16 [10] Sasaki (2021) UC Santa Cruz Diss.

[P] Anonymized pre-registration and materials for this experiment can be viewed at:
https://osf.io/dvshq/overview?view_only=8017742431b942caa67041a87bce0c68